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Physics

ENTHUSIAST COURSE



STUDY MATERIAL

Sound Wave

English Medium

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SOUND WAVE

KEY CONCEPT

Sound is a mechanical three dimensional and longitudinal wave that is created by a vibrating source such as a guitar string, the human vocal cords, the prongs of a tuning fork or the diaphragm of a loudspeaker.

1.0 EQUATION OF SOUND WAVES

As the piston oscillates sinusoidally, regions of compression and rarefaction are continuously set up. The distance between two successive compressions (or two successive rarefactions) equals the wavelength λ. As these regions travel through the tube, any small element of the medium moves with simple harmonic motion parallel to the direction of the wave. If s(x, t) is the position of a small element relative to its equilibrium position, We can express this harmonic position function as

$$s(x, t) = s_{max} \cos(kx - \omega t)$$

Where s_{max} is the maximum position of the element relative to equilibrium. This is often called the displacement amplitude of the wave. The parameter k is the wave number and ω is the angular frequency of the piston. Note that the displacement of the element is along s, in the direction of propagation of the sound wave, which means we are describing a longitudinal wave.

Consider a thin disk-shaped element of gas whose circular cross section is parallel to the piston in figure. This element will undergo changes in position, pressure, and density as a sound wave propagates through the gas. From the definition of bulk modulus, the pressure variation in the gas is

$$\Delta P = -B\frac{\Delta V}{V_i}$$

The element has a thickness Δx in the horizontal direction and a cross-sectional area A, so its volume is $V_i = A\Delta x$. The change in volume ΔV accompanying the pressure change is equal to $A\Delta s$, where Δs is the difference between the value of s at $x + \Delta x$ and the value of s at x. Hence, we can express ΔP as

$$\Delta P = -B\frac{\Delta V}{V_i} = -B\frac{A\Delta s}{A\Delta x} = -B\frac{\Delta s}{\Delta x}$$

As Δx approaches zero , the ratio $\Delta s/\Delta x$ becomes $\partial s/\partial x$ (The partial derivative indicates that we are interested in the variation of s with position at a fixed time.) Therefore,

$$\Delta P = -B\frac{\partial s}{\partial x}$$

If the position function is the simple sinusoidal function given by Equation , we find that

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$$\Delta P = -B \frac{\partial}{\partial x}[s_{max} \cos(kx - \omega t)] = Bs_{max} k \sin(kx - \omega t)$$

$$\Delta P = P_0 \sin(kx - \omega t)$$

$$P_0 = \frac{B\omega s_0}{v} = BKs_0$$

(B = Bulk modulus of the medium, K = angular wave number)

The displacement of a medium particle and excess pressure at any position are out of phase by $\frac{\pi}{2}$.

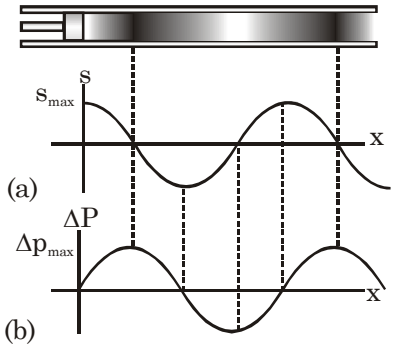


Figure : (a) Displacement amplitude and (b) pressure amplitude versus position for a sinusoidal longitudinal wave

2.0 SPEED OF SOUND WAVES

Velocity of sound waves in a linear solid medium is given by

$$v = \sqrt{\frac{Y}{\rho}} \qquad \text{(i)}$$

where Y = young's modulus of elasticity and ρ = density

Velocity of sound wave in a fluid medium (liquid or gas) is given by

$$v = \sqrt{\frac{B}{\rho}} \qquad \text{(ii)}$$

where, ρ = density of the medium and B = Bulk modulus of the medium given by,

$$B = -V \frac{dP}{dV} \qquad \text{(iii)}$$

Newton's formula : Newton assumed propagation of sound through a gaseous medium to be an isothermal process.

$$PV = \text{constant}$$

$$\Rightarrow \quad \frac{dP}{dV} = \frac{-P}{V}$$

and hence B = P

and thus he obtained for velocity of sound in a gas.

$$v = \sqrt{\frac{P}{\rho}} = \sqrt{\frac{RT}{M}} \text{ where M = molar mass}$$

Laplace's correction : Later Laplace established that propagation of sound in a gas is not an isothermal but an adiabatic process and hence $PV' = \text{constant}$

$$\Rightarrow \quad \frac{dP}{dV} = -\gamma \frac{P}{V}$$

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where, $B = -V\dfrac{dP}{dV} = \gamma P$

and hence speed of sound in a gas,

$$v = \sqrt{\frac{\gamma P}{\rho}} = \sqrt{\frac{\gamma RT}{M}} \qquad \text{(iv)}$$

2.1 Factors affecting speed of sound in atmosphere

(a) Effect of temperature : as temperature (T) increases velocity (v) increases. For small change in temperature above room temperature v increases linearly by 0.6 m/s for every 1°C rise in temp.

(b) Effect of humidity : With increase in humidity density decreases. This is because the molar mass of water vapour is less than the molar mass of air.

(c) Effect of pressure :

The speed of sound in a gas is given by $v = \sqrt{\dfrac{\gamma P}{P}} = \sqrt{\dfrac{\gamma RT}{M}}$

So at constant temperature, if P changes then ρ also changes in such a way that P/ρ remains constant. Hence pressure does not have any effect on velocity of sound as long as temperature is constant.

3.0 INTENSITY OF SOUND WAVES

Like any other progressive wave, sound waves also carry energy from one point of space to the other. This energy can be used to work, for example, forcing the eardrums to vibrate or in the extreme case of a sonic boom created by supersonic jet, can even cause glass panes of windows to crack.
The amount of energy carried per unit by a wave is called its power and power per unit area held perpendicular a sound wave travelling along positive x-axis described by the equation.

We define the intensity I of a wave, or the power per unit area , to be the rate at which the energy being transported by the wave transfers through a unit area A perpendicular to the direction of travel of the wave :

$$I = \frac{P}{A}$$

In the present case, therefore, the intensity is

$$I = \frac{P}{A} = \frac{1}{2}\rho A v(\omega s_{max})^2$$

Thus, we see that the intensity of a periodic sound wave is proportional to the square of the displacement amplitude and to the square of the angular frequency (as in the case of a periodic string wave). this can also be written in terms of the pressure amplitude ΔP_{max} ; in this case, we use Equation to obtain

$$I = \frac{\Delta P_{max}^2}{2\rho v}$$

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4.0 APPEARANCE OF SOUND TO HUMAN EAR

Pitch and Frequency

Frequency as we have discussed till now is an objective property measured its units of Hz and which can be assigned a unique value. However a person's perception of frequency is subjective. The brain interprets frequency primarily in terms of a subjective quality called Pitch. A pure note of high frequency is interpreted as high-pitched sound and a pure note of low frequency as low-pitched sound.

Pitch of a sound is that sensation by which we differentiate a buffalo voice, a male voice is of low pitch, a male voice has higher pitch and a female voice has (generally) still higher pitch. This sensation primarily depends on the dominant frequency present in the sound. Higher the frequency, higher will be the pitch and vice versa.

Loudness and Intensity

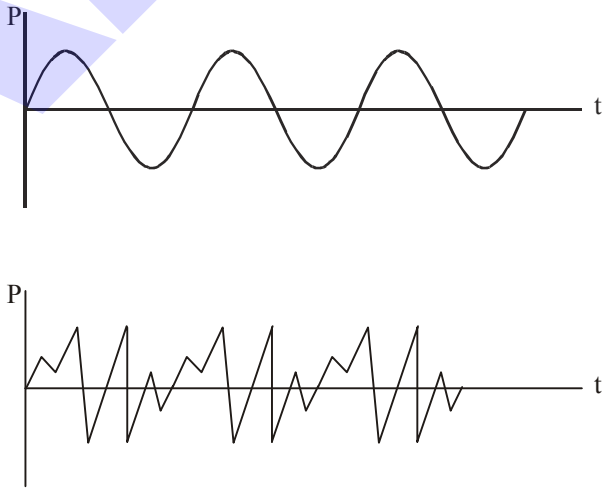
The loudness that we sense is related to the intensity of sound through it is not directly proportional to it. Our perception of loudness is better correlated with the sound level measured in decibels (abbreviated and dB) and defined as follows.

$$\beta = 10 \log_{10} \left(\frac{I}{I_0} \right),$$

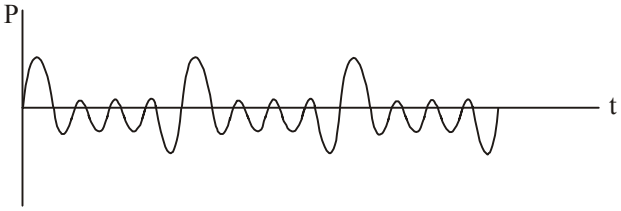
where I is the intensity of the sound and I_0 is a constant reference intensity 10^{-12} W/m^2. The reference intensity represents roughly the minimum intensity that is just audible at intermediate frequencies. For $I = I_0$, the sound level $\beta = 0$. Table shows the approximate sound levels of some of the sounds commonly encountered.

Quality and Waveform

A sound generated by a source may contain a number of frequency components in it. Different frequency components have different amplitudes and superposition of them results in the actual waveform. The appearance of sound depends on this waveform apart from the dominant frequency and intensity. Figure shows waveforms for a tuning fork, a clarinet and a cornet playing the same note (fundamental frequency = 440 Hz) with equal loudness.



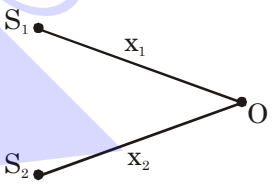
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We differentiate between the sound from a table and that from a mridang by saying that they have different quality. A musical sound has certain well-defined frequencies which have considerable amplitude. These frequencies are generally harmonics of a fundamental frequency. Such a sound is particularly pleasant to the ear. On the other hand, a noise has frequencies that do not bear well-defined relationship among themselves.

5.0 INTERFERENCE OF SOUND WAVES

If $p_1 = p_{m1} \sin(\omega t - kx_1 + \theta_1)$
and $p_2 = p_{m2} \sin(\omega t - kx_2 + \theta_1)$
resultant excess pressure at point O is



$p = p_1 + p_2$
$\Rightarrow p = p_0 \sin(\omega t - kx + \theta)$

where, $p_0 = \sqrt{p_{m_1}^2 + p_{m_2}^2 + 2p_{m_1}p_{m_2}\cos\phi}$, $\phi = |k(x_1 - x_2) + (\theta_1 - \theta_2)|$ ... (i)

(i) for constructive interference
$\phi = 2n\pi \quad \Rightarrow p_0 = p_{m1} + p_{m2}$
(ii) for destructive interference
$\phi = (2n + 1)\pi \quad \Rightarrow p_0 = |p_{m1} - p_{m2}|$

If ϕ is only due to path difference, then $\phi = \frac{2\pi}{\lambda}\Delta x$, and

Condition for constructive interference : $\Delta x = n\lambda$, $n = 0, \pm 1, \pm 2$

Condition for destructive interference : $\Delta x = (2n + 1)\frac{\lambda}{2}$, $n = 0 \pm 1, \pm 2$

from equation (6.1)

$P_0^2 = P_{m_1}^2 + P_{m_2} + 2P_{m_1} P_{m_2} \cos\phi$

Since intensity, $I \propto$ (Pressure amplitude)^2,

we have, for resultant intensity, $I = I_1 + I_2 + 2\sqrt{I_2 I_2}\cos\phi$ ... (ii)

If $I_1 = I_2 = I_0$
$I = 2I_0(1 + \cos\phi)$

$\Rightarrow I = 4I_0 \cos^2\frac{\phi}{2}$ ... (ii)

Hence in this case,
for constructive interference : $\phi = 0, 2\pi, 4\pi$ ..... and $I_{max} = 4I_0$
and for destructive interference : $\phi = \pi, 3\pi$ .... and $I_{min} = 0$

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Coherence : Two sources are said to be coherent if the phase difference between them does not change with time. In this case their resultant intensity at any point in space remains constant with time. Two independent sources of sound are generally incoherent in nature, i.e. phase difference between them changes with time and hence the resultant intensity due to them at any point in space changes with time.

6.0 LONGITUDINAL STANDING WAVES

Two longitudinal waves of same frequency and amplitude travelling in opposite directions interfere to produce a standing wave.

If the two interfering wave are given by

$p_1 = p_0 \sin(\omega t - kx)$

and $p_2 = p_0 \sin(\omega t + kx + \phi)$

then the equation of the resultant standing wave would be given by

$$p + p_1 + p_2 = 2p_0 \cos\left(kx + \frac{\phi}{2}\right) \sin\left(\omega t + \frac{\phi}{2}\right)$$

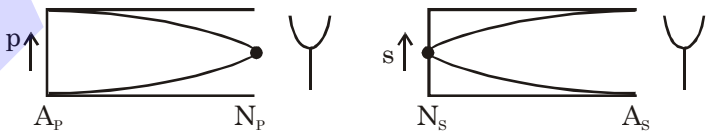
$$\Rightarrow \quad p = p_0' \sin\left(\omega t + \frac{\phi}{2}\right) \qquad \ldots (i)$$

6.1 Vibration of air columns

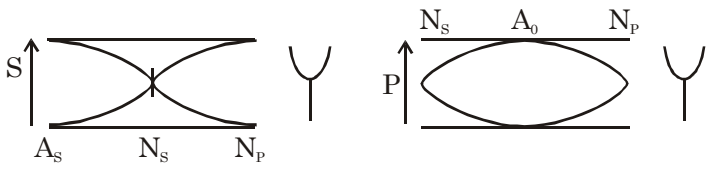
Standing waves can be set up in air-columns trapped inside cylindrical tubes if frequency of the tuning fork sounding the air column matches one of the natural frequency of air columns. In such a case the sound of the tuning fork becomes markedly louder, and we say there is resonance between the tuning fork and air column. To determine the natural frequency of the air column, notice that there is a displacement node (pressure antinode) at each closed end of the tube as air molecules there are not free to move, and a displacement antinode (pressure-node) at each open end of the air-column. In reality antinodes do not occurs exactly at the open end but a little distance outside. However if diameter of tube is small compared to its length, this end correction can be neglected.

Closed organ pipe

(In the diagram, A_P = Pressure antinode, A_S = displacement antinode, N_P = pressure node, N_s = displacement node.)



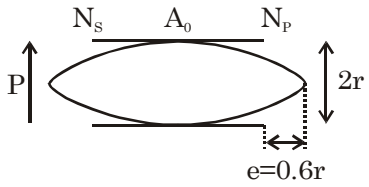
Open organ pipe :



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End correction : As mentioned earlier the displacement antinode at an open end of an organ pipe lies slightly outside the open lend. The distance of the antinode from the open end is called end correction and its value is given by

$e = 0.6\ r$



where r = radius of the organ pipe.

with end correction, the fundamental frequency of a closed pipe (f_c) and an open argon pipe (f_0) will be given by

$$f_c = \frac{v}{4(\ell + 0.6r)} \text{ and}$$

$$f_0 = \frac{v}{2(\ell + 1.2r)} \quad \text{..... (ii)}$$

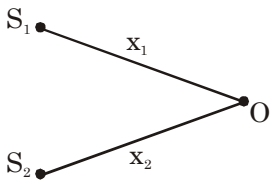
7.0 BEATS

When two sound waves of same amplitude and different frequency superimpose, then intensity at any point in space varies periodically with time. This effect is called beats.

If the equation of the two interfering sound waves emitted by s_1 and s_2 at point O are,

$p_1 = p_0 \sin(2\pi f_1 t - kx_1 + \theta_1)$

$p_2 = p_0 \sin(2\pi f_2 t - kx_2 + \theta_2)$



By principle of superposition

$p = p_1 + p_2$

$$= 2\pi_0 \cos\{\pi(f_1 - f_2)t + \frac{\theta_1 + \theta_2}{2}\} \sin\{\pi(f_1 + f_2)t + \frac{\theta_1 + \theta_2}{2}\}$$

i.e., the resultant sound at point O has frequency $\left(\frac{f_1 + f_2}{2}\right)$ while pressure amplitude $p'_0(t)$ varies with time as

$$p_0(t) = 2p_0 \cos\left\{\pi(f_1 - f_2)t + \frac{\phi_1 + \phi_2}{2}\right\}$$

Hence pressure amplitude at point O varies with time with a frequency of $\left(\frac{f_1 - f_2}{2}\right)$.

Hence sound intensity will vary with a frequency $f_1 - f_2$.

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This frequency is called beat frequency (f_B) and the time interval between two successive intensity maxima (or minima) is called beat time period (T_B)

$$f_B = f_1 - f_2$$

$$T_B = \frac{1}{f_1 - f_2}$$

8.0 DOPPLER EFFECT

(i) When a source of sound moves with a velocity v_s in a certain direction, the wavelength decreases in front of the source and increases behind the source.

$$\lambda' \text{ (in front)} = \frac{v - v_s}{f_s}\,;\ f' = \frac{v}{\lambda'} = \frac{v}{v - v_s} f_s$$

$$\lambda'' \text{(behind)} = \frac{v + v_s}{f_s}\,;\ f'' = \frac{v}{\lambda''} = \frac{v}{v + v_s} f_s$$

Here v is the velocity of sound in air.

(ii) The apparent frequency $= \frac{v - v_0}{v} f_s$

(a) When the source is moving towards the observer and the observer is moving away from the source, the apparent frequency

$$f_a = \frac{v - v_0}{v - v_s} f_s$$

$v_0 \leftarrow$ o $\quad$ $v_s \leftarrow$ s

(b) When the source and the observer are moving towards each other.

$$f_a = \frac{v + v_0}{v - v_s} f_s$$

o $\rightarrow v_0 \quad v_s \leftarrow$ s

(c) When the source and observer are moving away from each other,

$$f_a = \frac{v - v_0}{v + v_s} f_s$$

$v_0 \leftarrow$ o $\quad$ s $\rightarrow v_s$

(d) When the source is moving away from the observer and the observer is moving towards the source

$$f_a = \frac{v + v_0}{v + v_s} f_s$$

o $\rightarrow v_0 \quad$ s $\rightarrow v_s$

Here all velocities are relation to the medium.

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